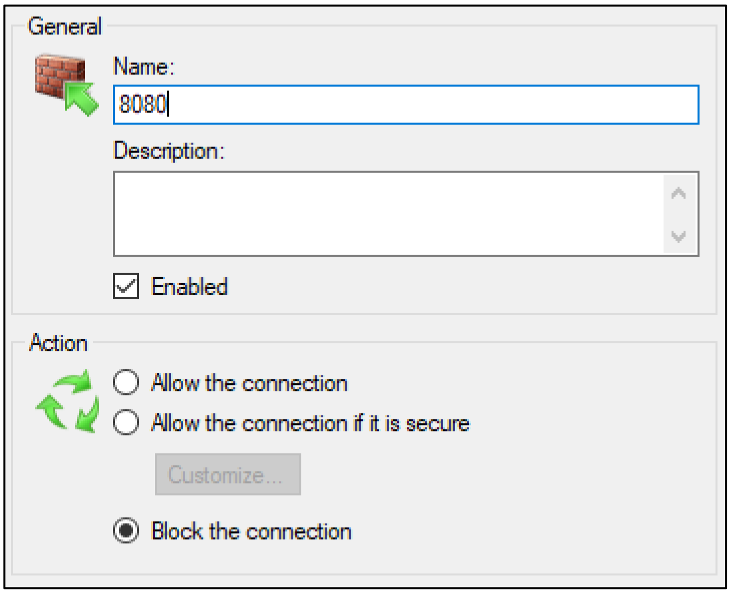
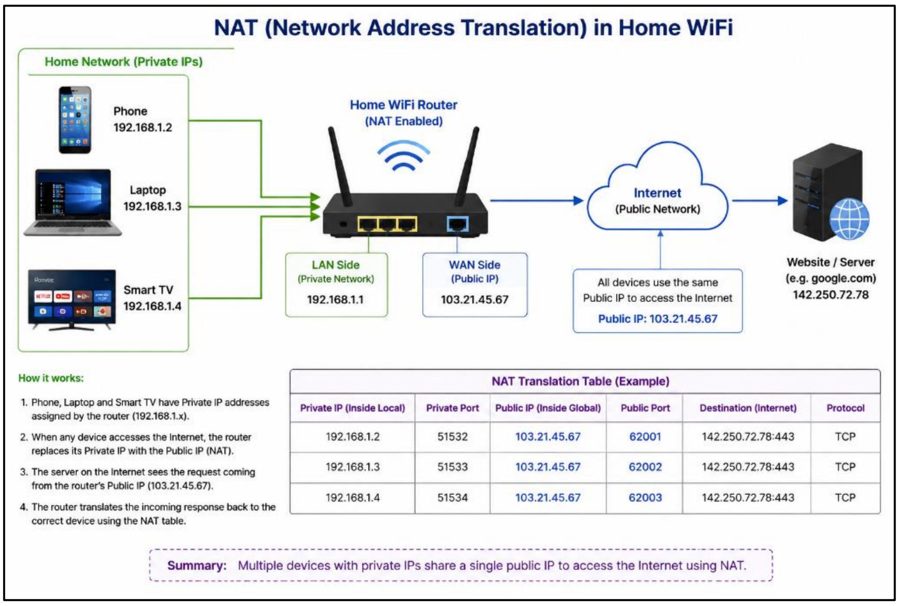
1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.



2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.



Private IP (Inside Local)	Private Port	Public IP (Inside Global)	Public Port	Destination (Internet)	Protocol
192.168.1.2	51532	103.21.45.67	62001	142.250.72.78:443	TCP
192.168.1.3	51533	103.21.45.67	62002	142.250.72.78:443	TCP
192.168.1.4	51534	103.21.45.67	62003	142.250.72.78:443	TCP

3. Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

1. Connect your laptop/PC to the Wi-Fi router.
2. Open a browser and enter the router's gateway address, commonly 192.168.1.1 or 192.168.0.1.
3. Log in using the router's administrator username and password.
4. Look for **Advanced Settings → Firewall → Port Filtering** (the menu name may vary by router).
5. Click **Add/New Rule**.

- Configure the rule:
- **Rule Name:** Block SMTP Port 25
- **Direction:** Outbound
- **Protocol:** TCP
- **Destination Port:** 25
- **Action: Block/Deny**
- **Source IP:** Any/All devices

6. Click **Save/Apply**.
7. Restart the router if it asks you to do so.

Current Filter Table

Select	Direction	Protocol	Source IP Address	Source Port	Destination IP Address	Destination Port	Interface	Rule Action
☐	Outgoing	TCP	192.168.1.50/32	25	203.0.113.5/32	443	any	Deny

4. Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.

- Firewalls evaluate rules sequentially (usually from top to bottom) and apply the first rule that matches the packet's source IP, destination IP, destination port, and protocol.
- To give you the exact "allowed" or "blocked" status for your three packets, please provide the rule set. It typically looks something like this:

- Rule 1: Allow Source 192.168.1.0/24 to Any Destination Port 80
- Rule 2: Block Source Any to Any Destination Port 22
- Default Rule: Deny All

5. Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

Ways Instagram or Zomato Could Use Firewalls and Packet Filtering

1. **Block unauthorized ports and services**
   Instagram or Zomato can use firewalls to allow only required ports, such as HTTPS (443), and block unnecessary ports. This reduces the number of services attackers can try to exploit.
2. **Filter suspicious or malicious traffic**
   Packet-filtering rules can block traffic from known malicious IP addresses, unusual source networks, or traffic that matches attack patterns. This can help reduce attacks such as scanning, brute-force attempts, and some forms of automated abuse.
3. **Control and limit excessive traffic**
   Firewalls can work with rate-limiting and filtering systems to restrict unusually large numbers of requests from a source. This helps protect servers from DDoS attacks and other traffic floods that could make Instagram or Zomato unavailable.